

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

End-to-End Data Analyst Project Report & GitHub README

Tools: Microsoft Excel | SQL/MySQL | Python/Pandas | Power Query | Power BI | DAX

1. Project Overview

Customer Shopping Behavior Analysis is an end-to-end data analytics project for understanding purchasing patterns, product performance, customer behavior, payment preferences, seasonal demand, subscriptions, discounts, and shipping choices.

The workflow covers data validation, cleaning, SQL analysis, Python exploratory analysis, and Power BI dashboard development.

2. Business Problem

A retail business has transaction data but needs a clear way to understand where sales come from, which products and categories perform best, how customers pay, which seasons are stronger, and how subscription, discount, and purchase-frequency behavior relate to purchases.

Key questions: Which products and categories generate the most sales? Which locations contribute most? Which season performs best? Which payment method is strongest? How do subscriptions and discounts compare?

3. Project Objectives

- Clean and validate transaction data.
- Use SQL for business analysis.
- Use Python and Pandas for EDA.
- Build Power BI measures with DAX.
- Create KPI cards, charts, and slicers.
- Generate business insights and recommendations.
- Demonstrate an end-to-end Data Analyst workflow.

4. Dataset Summary

Transactions: 3,900 | Columns: 19 | Distinct customers: 3,900 | Total sales: \$233,081.00 | Average purchase: \$59.76 | Average rating: 3.75/5 | Missing review ratings: 37

5. Dataset Columns

- Customer_ID - unique customer identifier
- Customer_Age - customer age
- Gender - customer gender
- Product_Name - purchased product
- Product_Category - product category
- Purchase_Amount_USD - purchase value / sales metric
- Customer_Location - customer location
- Product_Size - selected size
- Product_Color - selected color
- Purchase_Season - season
- Review_Rating - customer rating
- Subscription_Status - subscription status
- Shipping_Type - shipping method
- Discount_Applied - discount flag
- Promo_Code_Used - promo-code flag
- Previous_Purchases - previous purchase count
- Payment_Method - payment method

- Purchase_Frequency - purchase frequency
- Purchase_Count - one transaction per row

6. Important Data Limitation

The supplied dataset does not contain Order_Date, Quantity, Customer_Name, or Order_ID. Therefore, monthly/yearly sales trends and unit-volume analysis should not be claimed. This limitation is documented in the supplied project guide.

7. End-to-End Workflow

Raw Data → Excel/Power Query Cleaning → SQL Analysis → Python/Pandas EDA → Power BI Model → DAX → Dashboard → Insights

8. Excel / Power Query Cleaning

- Check data types and column names.
- Check and remove duplicates where appropriate.
- Identify missing values, especially Review_Rating.
- Trim text values.
- Validate age, purchase amount, rating, and previous purchases.
- Use appropriate numeric and currency formats.

9. SQL Analysis

Example - Total Sales:

```
SELECT SUM(Purchase_Amount_USD) AS Total_Sales FROM customer_shopping;
```

Example - Sales by Category:

```
SELECT Product_Category, SUM(Purchase_Amount_USD) AS Total_Sales FROM customer_shopping GROUP BY Product_Category ORDER BY Total_Sales DESC;
```

Example - Top Customers:

```
SELECT Customer_ID, SUM(Purchase_Amount_USD) AS Total_Sales FROM customer_shopping GROUP BY Customer_ID ORDER BY Total_Sales DESC LIMIT 10;
```

10. Python / Pandas EDA

```
import pandas as pd
df = pd.read_csv('Customer_Shopping_Behavior_Analysis_Improved.csv')
print(df.shape) print(df.info()) print(df.isnull().sum()) print(df.describe())

category_sales =
df.groupby('Product_Category')['Purchase_Amount_USD'].sum().sort_values(ascending=False)
```

11. Power BI Data Model

The current dataset can be analyzed with a single Customer_Data table. If the business adds more sources, a star schema can be introduced with Customer, Product, Date, and Location dimensions.

12. DAX Measures

Total Sales = SUM('Customer_Data'[Purchase_Amount_USD])

Total Purchases = SUM('Customer_Data'[Purchase_Count])

Total Customers = DISTINCTCOUNT('Customer_Data'[Customer_ID])

Average Purchase = AVERAGE('Customer_Data'[Purchase_Amount_USD])

Average Rating = AVERAGE('Customer_Data'[Review_Rating])

Sales Per Customer = DIVIDE([Total Sales], [Total Customers])

13. Dashboard Design

KPI Cards: Total Sales, Total Purchases, Total Customers, Average Purchase, Average Rating.

Charts: Sales by Category, Top Products, Sales by Location, Sales by Season, Payment Method, Subscription Status, Discount Analysis.

Slicers: Gender, Product Category, Location, Season, Size, Subscription, Payment Method, Shipping Type, Discount, Purchase Frequency.

14. Business Questions

- What is total sales?
- What is average purchase value?
- Which category generates highest sales?
- Which product generates highest sales?
- Which location contributes most sales?
- Which season performs best?
- Which payment method contributes most sales?
- How do subscribers compare with non-subscribers?
- How does discount usage compare with purchase value?

15. Initial Dataset Findings

Highest-sales category: Clothing

Highest-sales product: Blouse

Highest-sales season: Fall

Highest-sales payment method: Credit Card

Average purchase: \$59.76

Average rating: 3.75/5

These are descriptive findings from the supplied dataset and should not be presented as causal conclusions.

16. Business Insight Format

Use: Observation → Business Meaning → Recommendation. Example: a high-revenue category indicates strong observed customer spending in that category; maintain availability and evaluate targeted promotions.

17. Recommendations

- Prioritize high-revenue categories and products.
- Monitor top products for stock availability.
- Use location analysis to identify strong and weak markets.
- Compare subscriber behavior before retention campaigns.
- Evaluate discounts using purchase count and average purchase value together.
- Use review ratings to identify products needing quality attention.
- Add transaction dates in future versions for trend analysis.

18. Skills Demonstrated

Excel | SQL | MySQL | Python | Pandas | NumPy | Matplotlib | Power Query | Power BI | DAX | Data Cleaning | EDA | Dashboard Development | Data Visualization | Business Insights

19. One-Minute Interview Answer

I developed a Customer Shopping Behavior Analysis project using Excel, SQL, Python, Pandas, Power Query, and Power BI. The dataset contains 3,900 purchase transactions. I cleaned and validated the data, used SQL for business queries, used Python/Pandas for exploratory analysis, and created a Power BI dashboard with DAX measures, KPI cards, charts, and slicers. The dashboard analyzes sales, products, categories, locations, seasons, payment methods, subscriptions, and customer behavior, and

converts the analysis into business insights.

20. GitHub README

Customer Shopping Behavior Analysis

End-to-end Data Analyst project using Excel, SQL, Python/Pandas, Power Query, Power BI and DAX.

Workflow: Data → Cleaning → SQL → Python EDA → Power BI → DAX → Dashboard → Insights.

Main metric: Purchase_Amount_USD. Dataset: 3,900 transactions and 19 fields.

Important limitation: no Order_Date or Quantity, so time-series and unit-volume analysis are not included.

21. Future Enhancements

- Add Order_Date for monthly and yearly trends.
- Add Quantity for unit-level analysis.
- Build Customer/Product/Date dimension tables.
- Add RFM customer segmentation.
- Add sales forecasting.
- Add machine-learning based customer prediction.
- Publish to Power BI Service with refresh when a live source is available.

22. Conclusion

This project demonstrates how raw transaction data can be transformed into useful business reporting using SQL, Python, Excel, Power Query, Power BI, and DAX. It provides a strong portfolio project for Data Analyst Trainee and Junior Data Analyst interviews.